Chromatographic investigations of macromolecules in the critical range of liquid chromatography, 5[a)]

Characterization of block copolymers of decyl and methyl methacrylate

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SUMMARY:

Block copolymers of decyl and methyl methacrylate were analyzed by liquid chromatography at the critical point of adsorption. The molar mass and the polydispersity of the decyl methacrylate block was determined using chromatographic conditions resulting in a critical mode for poly(methyl methacrylate). In addition, using the same column system, the total molar mass and polydispersity of the block copolymers were determined, operating at chromatographic conditions insuring a size exclusion mechanism for both components.

Introduction

The development and application of new polymeric materials with a complex structure and taylor-made properties has created new, demanding tasks for polymer analysis. As for copolymers, the determination of the molar mass distribution is not sufficient for a complete analysis because the distribution in chemical composition has to be considered as well.

Gradient elution or precipitation chromatography has been shown to be a very useful technique for the determination of the chemical heterogeneity of copolymers[1–3]. Chromatographic cross-fractionation, i.e. the combination of two different chromatographic techniques, is able to give detailed information on the molar mass distribution and the chemical heterogeneity[3–5].

Another more recent approach to the characterization of heteropolymers is the concept of "invisibility", which assumes that chromatographic conditions exist under which heteropolymers may be separated according to the size of one of the components, because the other component is chromatographically "invisible"[6,7]. This concept experimentally relates to liquid chromatography at the critical point of adsorption, which was developed by Entelis, Evreinov and Gorshkov as a method for

a) Part 4: H. Pasch, C. Brinkmann, Y. Gallot, *Polymer*, in press.

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the determination of the functionality-type distribution of telechelic oligomers and polymers[8,9].

It was shown by us[10,11] and other authors[12,13] that chromatography at the critical point of adsorption can be applied to the characterization of block copolymers. Taking a block copolymer A_nB_m, the block A_n may be regarded as a functional group or inhomogeneity. Therefore, using the critical conditions of B_m for the chromatographic separation, A_n may be analyzed, and vice versa.

The present paper describes the analysis of block copolymers of decyl and methyl methacrylate (DMA and MMA). Operating at the critical point of PMMA the molar mass and polydispersity of the DMA block shall be determined.

Experimental part

The block copolymers were prepared by group transfer polymerization as was described in a previous report[14]. In all cases poly(decyl methacrylate) constituted the first block.

The liquid chromatographic investigations were conducted on a modular HPLC system, comprising a Waters Model 510 pump, a Waters differential refractometer, a Knauer UV/VIS filter photometer, a Rheodyne six-port injection valve and a Waters column oven. Two different column systems were used, (a) a Macherey-Nagel Nucleosil Si-100 column, 200 × 4 mm inner diameter, with an average particle size of 5 μm, and (b) an in-line combination of two columns Merck LiChrospher Si-300 and Si-1000, 200 × 4 mm inner diameter, with an average particle size of 10 μm.

The flow rate was 0,5 mL/min, 20 μL of 0,5 – 1 wt.-% polymer solutions were injected. The solvents methyl ethyl ketone and cyclohexane were Baker HPLC grade.

The column temperature was kept at 25 °C for all experiments.

Results and discussion

As was stated previously, in liquid chromatography of macromolecules three modes may be operating: size exclusion, critical and adsorption. Depending on the polarity of the stationary phase and the composition of the mobile phase, one of these modes always will direct the retention of a certain polymer type. Given a certain stationary phase, the composition of the mobile phase can be changed in such a way that the retention behaviour of the macromolecule changes from size exclusion to adsorption and vice versa. This change in retention behaviour is demonstrated in Fig. 1A for poly(methyl methacrylate). Using Nucleosil Si-100, normal size exclusion behaviour is obtained, when the mobile phase comprises methyl ethyl ketone. By adding cyclohexane to the mobile phase the negative slope of the calibration curve log M vs. retention time increases, indicating that the formerly pure size exclusion mechanism gradually changes to a mixed behaviour. It is well known from size exclusion chromatography (SEC) of polymers that the addition of a nonsolvent (cyclohexane) to a solvent (methyl ethyl ketone) causes later elution due to the decrease of the hydrodynamic volume. In addition, depending on the polarity of the stationary phase and the polymer sample, enthalpic interactions start to affect the retention behaviour. By further decreasing the solvent strength of the mobile phase these interactions increase until they become predominant. At a mobile-phase composition of 60 vol.-% methyl ethyl ketone and 40 vol.-% cyclohexane typical adsorption behaviour is obtained. The retention time

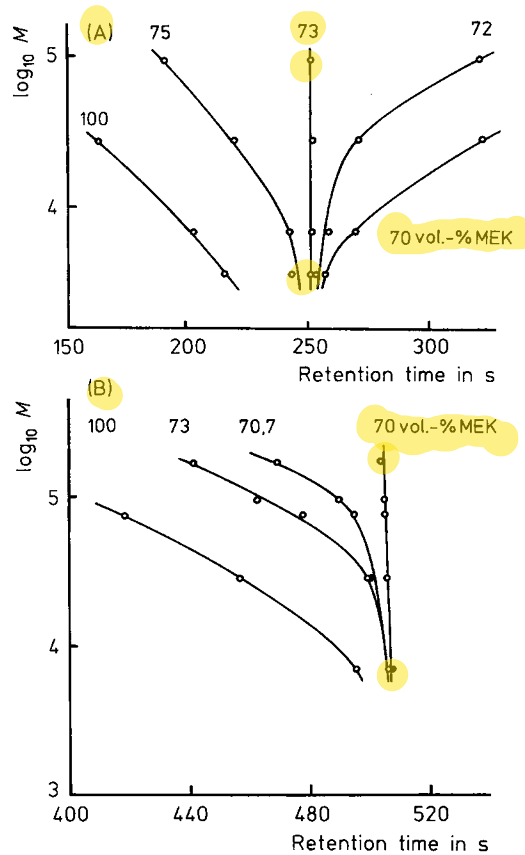

Fig. 1. Critical diagrams of log M vs. retention time; mobile phase: methyl ethyl ketone (MEK)-cyclohexane, stationary phase: Nucleosil Si-100 (A) or LiChrospher Si-300 + Si-1000 (B)

increases exponentially with increasing molar mass of the PMMA, and samples of higher molar mass irreversibly adsorb onto the stationary phase. The transition from size exclusion to adsorptions modes occurs at a mobile-phase composition of 73 vol.-% methyl ethyl ketone and 27 vol.-% cyclohexane. At this point, the critical point of adsorption of PMMA, entropic and enthalpic interactions between the macromolecules and the stationary phase exactly compensate each other so that, regardless of the chain length, all macromolecules are eluted at the same retention time. At the critical point of adsorption the distribution coefficient K_d equals 1 and the polymer chain does not contribute to retention.

As the elution of macromolecules in the critical mode is very sensitive to temperature[15], all experiments are conducted at a constant temperature of 25 °C.

The conditions, where the critical mode is operating, are affected by the composition of the mobile phase as well as the characteristics of the stationary phase. Depending on pore size and polarity of the stationary phase, the mobile-phase composition corresponding to the critical mode may change significantly even for allegedly similar stationary phases. As can be seeen in Fig. 1B, the critical point of PMMA is shifted from 73 vol.-% to 70 vol.-% methyl ethyl ketone in the mobile phase, when a column system comprising LiChrospher Si-300 and Si-1000 is used instead of Nucleosil Si-100. This effect must be taken into account, when columns similar in chemical structure but different in pore size are compared.

It was shown by previous investigations[16,17] that diblock copolymers may be characterized with respect to molar mass and molar mass distribution of one block, operating at the critical point of the other block. On the other hand, the molar mass of the whole block copolymer may be determined with the same stationary phase by changing the mobile phase to a composition corresponding to a pure size exclusion mode for both block components.

Therefore, in agreement with the behaviour of PMMA shown in Fig. 1, pure methyl ethyl ketone is used as the mobile phase for the determination of the molar mass of the block copolymers PDMA-*b*-PMMA, whereas the corresponding critical conditions of PMMA are used for the determination of the molar mass of the poly(decyl methacrylate) (PDMA) block.

The behaviour of the lowest molar mass block copolymer B1 in the size exclusion mode for both components (100% methyl ethyl ketone) on one hand and in the critical mode for MMA (73 vol.-% methyl ethyl ketone) on the other hand is shown in Fig. 2. At the critical point of PMMA the block copolymer is eluted in a size exclusion mode with respect to the PDMA block only, and typical SEC profiles are obtained.

A narrow molar mass distribution is to be expected for a polymer prepared by a living polymerization technique. Tab. 1 shows that the polydispersity of the block copolymer

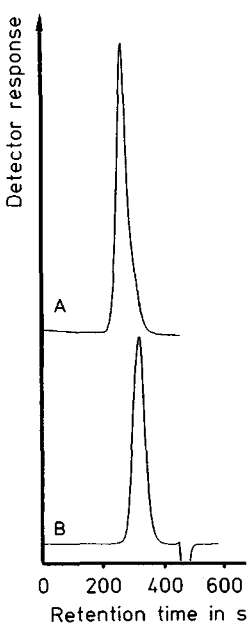

Fig. 2. Chromatograms of the block copolymer B1 in different chromatographic modes; stationary phase: Nucleosil Si-100, mobile phase: 100% methyl ethyl ketone (A) or methyl ethyl ketone-cyclohexane 73 : 27 v/v (B)

Tab. 1. Molar mass and polydispersity ($U = \bar{M}_w/\bar{M}_n - 1$) of the block copolymer B1, determined in different chromatographic modes, stationary phase Nucleosil Si-100

Sample	Monomer mole ratio DMA/MMA	Nominal			
		$\bar{M}_w$ (total)[a]	U (total)	$\bar{M}_w$ (DMA)[b]	
B1	25/75	59400	0,28	25500	
		Determined			
		$\bar{M}_w$ (total)[c]	U (total)[c]	$\bar{M}_w$ (DMA)[d]	U (DMA)[d]
		60300	0,24	16800	0,17

[a] Determined by conventional SEC[14].
[b] Calculated from the composition of the monomer mixture.
[c] Mobile phase 100% methyl ethyl ketone, calibration with PMMA.
[d] Mobile phase methyl ethyl ketone-cyclohexane 73:27 v/v, calibration with polystyrene.

is low in agreement with this expectation. The molar mass has been calculated using a PMMA calibration curve. As the calibration curves of both homopolymers, PMMA and PDMA, proved to be identical within the limitations of the method, the PMMA calibration is assumed to be valid for the copolymer as well.

At the critical point of PMMA, the PDMA block of the block copolymer alone ought to be in control of the elution behaviour and, therefore, a much lower molar mass is expected as compared to the previous experiment. And, indeed, good agreement is obtained between the molar mass determined thereby and the molar mass $\bar{M}_w$ (DMA) calculated from the composition of the monomer mixture. This is strong evidence for the fact that at the critical point of PMMA the elution behaviour is controlled by the PDMA block exclusively.

All other samples, higher in molar mass, were investigated on a column system comprising LiChrospher Si-300 and Si-1000 stationary phases. Similarly to the previous investigations, the total molar mass of the block copolymers was determined using pure methyl ethyl ketone as the mobile phase, thus substantiating a size exclusion mechanism for both components. The results, summarized in Tab. 2, demonstrate perfect agreement with the values determined by conventional SEC[14].

Tab. 2. Molar mass and polydispersity of the block copolymers B2–B4, stationary phase LiChrospher Si-300 + Si-1000, mobile phase 100% methyl ethyl ketone

Sample	Nominal[a]		Determined	
	$\bar{M}_w$ (total)	U (total)	$\bar{M}_w$ (total)	U (total)
B2	51200	0,13	55900	0,15
B3	121000	0,21	115300	0,19
B4	149000	0,21	146900	0,21

[a] Determined by conventional SEC[14].

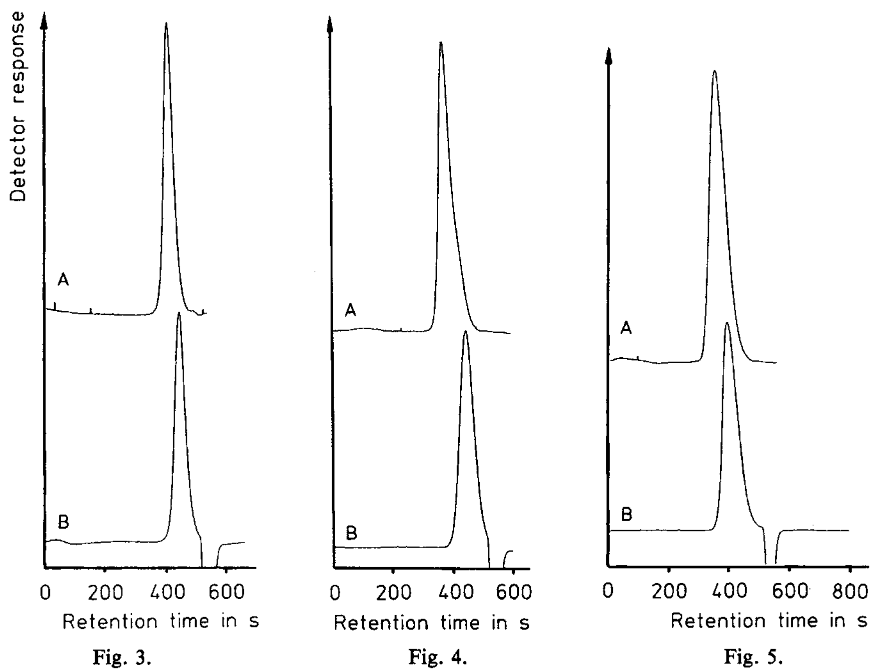

Fig. 3. Chromatograms of the block copolymer B2 in different chromatographic modes; stationary phase: LiChrospher Si-300 + Si-1000, mobile phase: 100% methyl ethyl ketone (A) or methyl ethyl ketone-cyclohexane 70:30 v/v (B)

Fig. 4. Chromatograms of the block copolymer B3 in different chromatographic modes; conditions as in Fig. 3

Fig. 5. Chromatograms of the block copolymer B4 in different chromatographic modes; conditions as in Fig. 3

The molar mass parameters of the PDMA block in the copolymers were determined by conducting the chromatographic separations at the critical point of PMMA. In the case of the two-column system Si-300 + Si-1000, the critical point of PMMA corresponds to a mobile phase composition of methyl ethyl ketone-cyclohexane 70:30 v/v (see Fig. 1). The chromatograms of the block copolymers in the size exclusion and critical modes are shown in Figs. 3–5. Again, at the critical point, the elution profiles correspond to lower molar masses.

Similar to conventional SEC, quantification is based on a calibration curve log M vs. retention time. Obviously, for the PDMA block a PDMA calibration curve would be most useful; corresponding calibration standards, however, are not commercially available. Unfortunately, PMMA cannot serve as calibration standard, because these experiments are carried out at the critical point of PMMA, so that — by definition — all PMMA samples are eluted at a single retention time regardless of their molar mass.

Tab. 3. Molar mass and polydispersity of the PDMA block in the block copolymers determined at the critical point of PMMA, stationary phase LiChrospher Si-300 + Si-1000, mobile phase methyl ethyl ketone-cyclohexane 70:30 v/v

Sample	Monomer mole ratio DMA/MMA	Nominal[a] $\overline{M}_w$ (DMA)	Determined			
			$\overline{M}_w$ (DMA)[b]	U (DMA)[b]	$\overline{M}_w$ (DMA)[c]	U (DMA)[c]
B2	75/25	44600	32300	0,19	46300	0,20
B3	25/75	52000	32800	0,30	46600	0,30
B4	50/50	103300	71000	0,26	87100	0,21

[a] Calculated from the total molar mass and the monomer ratio.
[b] Polystyrene calibration.
[c] Poly(decyl methacrylate) calibration.

Therefore, a polystyrene (PS) calibration curve was used; this yielded molar mass averages and polydispersities, which were in fair agreement with the nominal values calculated from the composition of the monomer mixtures (see Tab. 3). For comparison, a few samples of poly(decyl methacrylate) with broader molar mass distribution were analyzed, too. The corresponding calibration curve differed significantly from the PS calibration curve (see Fig. 6); it was used for a tentative quantification (see Tab. 3) of the PDMA blocks. As the calibration curves are parallel to each other, the polydispersity is not affected by the type of calibration. The $\overline{M}_w$ values, however, obtained by using the PDMA calibration, are gradually higher than the values based on the PS calibration. Fortunately, two of the samples under investigation (B3, B4) had been characterized by gradient HPLC in a previous study[14]. The following number-

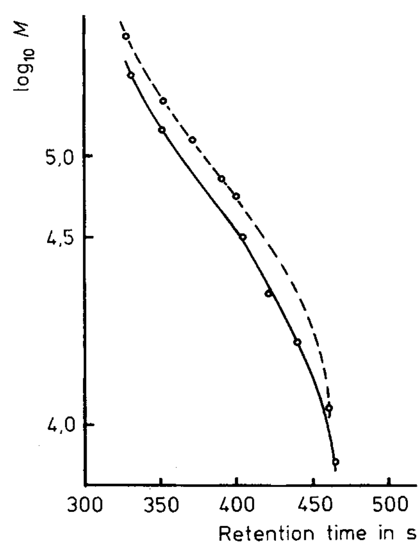

Fig. 6. Calibration curves: log M vs. retention time of polystyrene (———) and poly(decyl methacrylate) (– – –), stationary phase: LiChrospher Si-300 + Si-1000, mobile phase: methyl ethyl ketone-cyclohexane 70:30 v/v

average degrees of polymerization, P_n, were calculated from the results of that investigation, using a normal-phase solvent gradient of isooctane/tetrahydrofuran in combination with evaporative light scattering detection.

B3: P_n (DMA) = 190 $\bar{M}_n$ (DMA) = 42900

B4: P_n (DMA) = 380 $\bar{M}_n$ (DMA) = 85900

Within the limits of experimental error, these data are close to those calculated using the PDMA calibration. However, further efforts are necessary to improve the quantification procedure.

The determination of the molar mass distribution of the second block component, i.e. the PMMA block, would have to be carried out on a different chromatographic system. Analogous to the experiments described in this paper, the PMMA block was to be analyzed at the critical point of PDMA. Using these conditions, however, PMMA would be eluted in an adsorption mode from a silica gel stationary phase and irreversible adsorption would be likely to occur for high molar mass samples. Therefore, a reversed stationary phase such as RP-18 is more likely to be suited for an analysis of the PMMA block.

Conclusion

Liquid chromatography at the critical point of adsorption is a unique method for the characterization of block copolymers. It was shown for PDMA-*b*-PMMA that, operating at the critical point of the PMMA block, the block copolymer can be eluted in a size exclusion mode with respect to the PDMA block. Via an appropriate calibration, molar mass and polydispersity of the PDMA block can be determined. Additionally, the total molar mass of the block copolymer can be determined using the same stationary phase and chromatographic conditions corresponding to a size exclusion mode for both components.

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